

OSLAB PRACTICAL-2 PROGRAMS & OUTPUT

Date: 05-08-26

Program-1 & OUTPUT:

```
sainath@Sainath:~/practical_2$ cat program1.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("sample.txt", O_RDONLY);

    if (fd == -1) {
        printf("File cannot be opened.\n");
        return 1;
    }

    printf("File opened successfully.\n");

    close(fd);

    return 0;
}
sainath@Sainath:~/practical_2$ ./program1
File opened successfully.
```

Program-2 & OUTPUT:

```
sainath@Sainath:~/practical_2$ cat program2.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;
    char buffer[100];
    int bytes;

    fd = open("sample.txt", O_RDONLY);

    bytes = read(fd, buffer, sizeof(buffer)-1);

    buffer[bytes] = '\0';

    printf("File Content:\n%s", buffer);

    close(fd);

    return 0;
}
sainath@Sainath:~/practical_2$ ./program2
File Content:
Hello Operating Systems Lab
```

Program-3 & OUTPUT:

```
sainath@Sainath:~/practical_2$ cat program3.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main() {

    int fd;

    char text[] = "Operating Systems Lab";

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    write(fd, text, strlen(text));

    printf("Data written successfully.\n");

    close(fd);

    return 0;
}
sainath@Sainath:~/practical_2$ ./program3
Data written successfully.
```

Program-4 & OUTPUT:

```
sainath@Sainath:~/practical_2$ cat program4.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main() {

    int fd;

    fd = open("sample.txt", O_RDONLY);

    if(fd == -1) {
        printf("Cannot open file.\n");
        return 1;
    }

    printf("File opened.\n");

    close(fd);

    printf("File closed successfully.\n");

    return 0;
}
sainath@Sainath:~/practical_2$ ./program4
File opened.
File closed successfully.
```

history:

```
sainath@Sainath:~/practical_2$ history
```

```
1 nano demo.c
2 gcc demo.c -o demo
3 sudo apt update
4 gcc demo.c -o demo
5 sudo apt install build-essential
6 gcc demo.c -o demo
7 mkdir OS programs
8 mkdir OSLAB
9 cd OSLAB
10 mkdir fork
11 cd fork
12 gedit program1.c
13 apt update
14 apt install gedit
15 gedit --version
16 sudo snap install gedit
17 sudo apt install gedit
18 gedit program1.c
19 gcc program1.c -o process
20 ./program1
21 ./process
22 gedit program2.c
23 gcc program2.c -o process
24 ./process
25 gedit program3.c
26 sudo apt install gedit
27 gcc program3.c -o process
28 ./process
29 gedit program4.c
30 gcc program4.c -o process
31 ./process
32 gedit program5.c
33 gcc program5.c -o process
34 ./process
35 gedit program6.c
36 gcc program6.c -o process
37 ./process
38 gedit program7.c
39 gcc program7.c -o process
40 ./process
41 gedit program10.c
42 gcc program10.c -o process
43 ./process
44 gedit program9.c
45 gcc program9.c -o process
46 ./process
47 pwd
48 ~/ OSLAB
49 cd ~/OSLAB
50 ls
51 cd ~/fork
52 ls
53 cat program1.c
54 pwd
55 cd ~/ OSLAB
```

```
56 cd ~/OSLAB
57 LS
58 ls
59 cd ~/program1.c
60 cd ~/OSLAB
61 cd fork
62 ls
63 cd program1.c
64 cat program1.c
65 ./program1
66 ./program1.c
67 ./process
68 cd ~/OSLAB
69 cd fork
70 ls
71 cat program1.c
72 ./program1.c
73 ./program1
74 ./program1.c
75 ls
76 pwd
77 cd OSLAB
78 mkdir exec
79 cd exec
80 gedit program1.c
81 gedit program1.c
82 gcc program1.c
83 gcc program1.c -o program1.c
84 gcc program1.c -o program1
85 ./program1
86 gedit program2.c
87 gcc program2.c -o program2
88 ./program2
89 gedit program3.c
90 gcc program3.c -o program3
91 ./program3
92 gedit program4.c
93 gcc program4.c -o program4
94 ./program4
95 gedit program5.c
96 gcc program5.c -o program5
97 ./program5
98 cat program1.c
99 ./program1
100 cat program2.c
101 ./program2
102 cat program3.c
103 ./program3
104 cat program4.c
105 ./program4
106 cat program5.c
107 ./program5
108 cd fork
109 cd ~/OSLAB
110 cd fork
111 cat program1.c
```

```
112 ./program1
113 ./program1.c
114 cd ~/OSLAB
115 cd fork
116 cat fork
117 cd ~/OSLAB
118 cd fork
119 cat program1.c
120 gcc program1.c -o program1
121 ls -l
122 ./program1
123 cd ~/OSLAB
124 cd fork
125 cat program1.c
126 gcc program1.c -o program1
127 ./program1.c
128 ls -l
129 ./program1.c
130 ./program1
131 cd ~/OSLAB
132 cd fork
133 cat program1.c
134 gcc program1.c -o program1
135 ./program1
136 cat program2.c
137 ./program2
138 ./program2.c
139 cd ~/OSLAB
140 cd fork
141 cat program1.c
142 gcc program1.c -o program1
143 ./program1
144 cat program2.c
145 gcc program2.c -o program2
146 ./program2
147 cat program3.c
148 gcc program3.c -o program3
149 ./program3
150 cat program4.c
151 gcc program4.c -o program4
152 ./program4
153 cat program5.c
154 gcc program5.c -o program5
155 ./program5
156 cat program6.c
157 gcc program6.c -o program6
158 ./program6
159 cat program7.c
160 gcc program7.c -o program7
161 ./program7
162 cat program8.c
163 cat program9.c
164 gcc program9.c -o program9
165 ./program9
166 cat program10.c
167 gcc program10.c -o program10
```



```
168 ./program10
169 mkdir practical_1
170 cd practical_1
171 gedit program1.c
172 gcc promgram1.c -o program1
173 gcc program1.c -o program1
174 gcc program1.c -o program1
175 ./program1
176 gedit program2.c
177 gcc program2.c -o program2
178 ./program2
179 gedit program3.c
180 gcc program3.c -o program3
181 ./program3
182 gedit program4.c
183 gcc program4.c -o program4
184 ./program4
185 gedit program5.c
186 gcc program5.c -o program5
187 ./program5
188 cat program1.c
189 ./program1.c
190 ./program1
191 cat program1.c
192 ./program1
193 cat program2.c
194 ./program2
195 cat program3.c
196 ./program3
197 cat program4.c
198 ./program4
199 cat program5.c
200 ./program5
201 uname -a
202 lscpu
203 lsblk
204 ps
205 top
206 history
207 mkdir practical_2
208 cd practical_2
209 gedit program1.c
210 gcc program1.c -o program1
211 ./program1
212 echo "Hello Operating Systems Lab" > sample.txt
213 ls
214 ./program1
215 gedit program2.c
216 gcc program2.c -o program2
217 ./program2
218 gedit program3.c
219 gcc program3.c -o program3
220 ./program3
221 gedit program4.c
222 gcc program4.c -o program4
223 ./program4
```

```
223 ./program4
224 cat program1.c
225 ./program1
226 cat program2.c
227 ./program2
228 cat program3.c
229 ./program3
230 cat program4.c
231 ./program4
232 gcc 01_open.c -o open
233 gcc 02_read.c -o read
234 gcc 03_write.c -o write
235 gcc 04_close.c -o close
236 gcc 01_open.c -o open
237 history
```